Machine Learning Analysis Report: Breast Cancer Diagnosis Classification

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GitHub Repository: https://github.com/sandipdey2/breast-cancer-classification

Overview

This project implements a supervised binary classification model to predict whether a breast tumor is malignant or benign. The dataset used is the Breast Cancer Wisconsin (Diagnostic) dataset from the UCI Machine Learning Repository (https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic). A logistic regression model was trained on 30 numeric cell-nucleus features extracted from fine-needle aspirate images. The goal was to demonstrate a complete, interpretable machine learning workflow rather than maximize accuracy.

Dataset Description

The dataset contains 569 samples of breast tumor measurements and 32 columns after cleaning (ID, diagnosis, and 30 numeric features). Each sample is labeled as malignant (M) or benign (B). The 30 features describe mean, standard error, and worst values of ten cell characteristics such as radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, and fractal dimension. The class distribution is moderately imbalanced (approximately 357 benign and 212 malignant). No missing values remain after dropping an empty trailing column. The data are real clinical measurements and suitable for academic classification work.

Modeling Approach

The ID column was removed because it carries no predictive information. The target was label-encoded (malignant = 1, benign = 0). A stratified 75/25 train-test split preserved class proportions. Features were standardized with StandardScaler because logistic regression assumes comparable feature scales. Logistic regression was chosen for its simplicity, interpretability, and suitability to a binary classification task with continuous features (James et al., 2013). Accuracy, precision, recall, and F1-score were selected as evaluation metrics because the clinical cost of false negatives (missing a malignant case) is high, making recall especially relevant, while F1 balances precision and

recall under class imbalance (Saito & Rehmsmeier, 2015). The model assumes a linear decision boundary in the scaled feature space.

Results

On the held-out test set (143 samples: 90 benign, 53 malignant) the logistic regression model achieved the following performance:

- **Accuracy:** 0.9650
- **Precision:** 0.9800
- **Recall:** 0.9245
- **F1-score:** 0.9515

The classification report shows Benign precision 0.96 / recall 0.99 and Malignant precision 0.98 / recall 0.92. The confusion matrix (Figure 1 in the notebook) confirms that most cases were correctly classified, with relatively few false negatives. These results indicate that the linear model captured useful signal in the cell-nucleus features for this classic benchmark dataset.

Interpretation for a Non-Technical Audience

The model was able to correctly identify most cancerous and non-cancerous tumors in the test data. In practical terms, it rarely missed a malignant case (recall 0.9245) and rarely flagged a healthy case as cancerous (precision 0.9800). This does not mean the model is ready for clinical use; it only shows that the measured cell features contain useful information for distinguishing the two types of tumors in this particular dataset.

Limitations and Potential Bias

One limitation is the modest sample size (569 cases) and the fact that all data come from a single historical source, which may not represent current patient populations or imaging techniques. Another potential source of bias is that the original measurements were collected under specific clinical protocols; differences in equipment or patient demographics could reduce performance on new data. A practical step to reduce risk would be to evaluate the model on an external validation set or to use techniques such as cross-validation and calibration checks before any deployment consideration.

References

James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013). *An introduction to statistical learning*. Springer. https://doi.org/10.1007/978-1-4614-7138-7

Saito, T., & Rehmsmeier, M. (2015). The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on

imbalanced datasets. *PLOS ONE, 10*(3), e0118432. https://doi.org/10.1371/journal.pone.0118432